MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE DOD-005 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: TR tube with external shutter, bandpass, 15,800 to 17,000 MHz

ABSOLUTE RATINGS:

Parameter:	Es	Ii	Ignitor voltage	Transmitter po	TA	Du	Alt
Unit:	Vdc	μAdc	Vdc	kw	°C	---	ft
Maximum:	29	200	---	10	+125	0.001	60,000
Minimum:	21	100	-850	---	-55	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 5
Weight: 3.0 ounces (approx)
Mounting: Mates with UG-541/U choke flanges

TEST CONDITIONS:

Parameter:	Ignitor voltage	Transmitter po	tp	prr	F
Unit:	Vdc	kw	μs	pps	MHz
Test condition 1:					
Maximum:	870	---	---	---	---
	---	---	---	---	F1
Minimum:	850	---	---	---	---
Test condition 2:					
Maximum:	870	11	0.50	---	---
	---	---	---	2,400	F1
Minimum:	850	9	0.30	---	---
Test condition 3:					
Maximum	870	11	0.50	---	---
	---	---	---	2,400	F3
Minimum:	850	9	0.30	---	---

Frequency		
F	MHz	±
F1	15,800	16
F2	16,000	16
F3	16,200	16
F4	16,500	16
F5	17,000	16

GENERAL:

Qualifications - Required

1/ Formerly BTR-537

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	3	---					
4401	Ignitor ignition time	4	1		t	---	5.0	sec
---	Ignitor current	4	1		Ii	130	165	μAdc
4421	Ignitor interaction	---	1		ΔLi	---	0.1	dB
4416	Insertion loss	7	1	F = F1, F4, and F5	Li	---	0.6	dB
4416	Insertion loss	7	---	F = F1, F4, and F5 (shutter closed)	Li	25	---	dB
4473	Low-level VSWR	5, 7	1	F = F1, F4, and F5; σ' = 1.02 (max)	σ	---	1.4	---
4452	Leakage power (flat)	---	3		pf	---	50	mw
4452	Leakage energy (spike)	---	3		Ws	---	0.25	erg
4471	Recovery time	6	3		t	---	1.0	μs
4496	Firing power	18	1	F = F3 ±810 MHz	Po/Du	30	200	mw
---	Phase shift	8, 9, 10	1	See figure 6 F = F1 F = F4 F = F5	 Δϕ Δϕ Δϕ	 44.5 346.0 309.5	 47.5 349.0 312.5	 Degrees Degrees Degrees
---	Shutter current	11	---	Es = 28 ±0.5 Vdc	Is	---	230	mAdc
---	Shutter cycling	12	---	Es = 28 ±1 Vdc; prr = 10 Hz (max); 250 cycles	---	---	---	---
---	Ignitor current noise test	19	---		---	---	---	---
1027	Temperature cycling (nonoperating)	---	---	TA = -55° to +125°C	---	---	---	---
	Quality conformance inspection, part 2							
4452	Leakage power (flat)	---	3	TA = 125° C	pf	---	50	mw
4452	Leakage energy (spike)	---	3	TA = 125° C	Ws	---	0.25	erg
4471	Recovery time	6	3	TA = -55° C	t	---	1.0	μs
4496	Firing power	18	1	F = F3 ±810 MHz; TA = 125° C	Po/Du	30	200	mw
---	Noise figure change	---	1	F = F1 thru F5	ΔNF	---	0.9	dB
---	Ignitor current	---	1	TA = 125° C TA = -55° C	Ii Ii	130 130	190 190	μAdc μAdc
4460	Ignitor noise ratio	---	---	F = F1 thru F5	Nr	---	1.3	---
---	Shutter current	11	---	Es = 28 ±0.5 Vdc TA = 125° C TA = -55° C	 Is Is	 --- ---	 200 230	 mAdc mAdc

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Quality conformance inspection, part 2 - Continued							
---	Shutter cycling	12	---	Es = 28 ± 1 Vdc; prr = 10 Hz (max); TA = 125° C; 1,000 cycles	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	16, 17	3	TA = 125° C (shutter open)	t	1,600	---	hrs
---	Life-test end points:	---	---					
4471	Recovery time	6	3		t	---	2.0	μs
---	Crystal degradation	16, 17	---		ΔNF	---	1.0	dB
4416	Insertion loss	7	1	F = F1 thru F5	Li	---	0.6	dB
---	Noise figure change	---	---	F = F1 thru F5	ΔNF	---	1.0	dB
---	Phase shift	---	---	See figure 6 F = F1 F = F4 F = F5	 Δθ Δθ Δθ	 44.5 340.0 309.5	 47.5 349.0 312.5	 Degrees Degrees Degrees
4496	Firing power	18	1	F = F3 ±810 MHz	Po/Du	30	200	mw
4452	Leakage energy (spike)	---	2		Ws	---	0.25	erg
---	Shutter life test	12	---	Es = 29 $^{+1}_{-0}$ Vdc; prr = 10 Hz (max); TA = 125° C	--	50,000	---	Cycles
---	Shutter life-test end points:	---	---					
4416	Insertion loss	7	---	F = F1 thru F5 (shutter closed)	Li	25	---	dB
4416	Insertion loss	7	1	F = F1 thru F5	Li	---	0.6	dB
4021	Degradation due to vibration	---	---		---	---	---	---
---	Shutter phase modulation	14	---	F = F4; Es = 21 ± 0.4 Vdc; TA = +125° C and +25° C; t = 10 minutes (min)	Δθ	---	0.1	Degree
	Periodic-check tests							
1002	Barometric pressure, reduced	---	1		Alt	60,000	---	ft
4021	Degradation due to vibration	13	---	F = 5 to 55 Hz; DA = 0.08 inch; F = 55 to 500 Hz; 10 G; Es = 21 Vdc; t = 15 minutes	---	---	---	---
---	Shutter holding current	15	---	TA = +25° C TA = +125° C	Is Is	--- ---	178 178	mAdc mAdc

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NOTES:

1. The shutter provides a convenient method of opening or shorting a waveguide line at low-power levels (incident peak power less than 1 kw). It is not intended for applications involving the switching of high power and should remain either open or closed whenever high power is incident on the tube.

2. Unless otherwise specified, all low-level and high-level tests are to be made with the shutter open.

3. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

4. *A resistor, in series with the ignitor, is encapsulated as an integral part of the tube. It is* selected to provide a current of 150 μAdc when the specified voltage, Ebb = -860 Vdc, is applied to the ignitor terminal (No. 1) of the tube.

5. A swept frequency method of measurement may be used instead of measurement at fixed frequencies. The VSWR over the range of 15.80 to 16.50 GHz shall not be greated than 1.4 at any point.

6. The recovery time shall be measured at the point where the signal level has recovered to within 0.5 dB of the full signal level.

7. A swept frequency signal may be used for this test.

8. The phase length of the tube shall be measured relative to an equivalent length of RG-91/U waveguide.

9. When using a phase shifter (rotary vane type such as Hewlett Packard No. P885A, or equivalent) record the phase shifter dial reading using only the top scale and vernier, when the circuit has been balanced.

10. The phase slope deviation shall not exceed ±1.5 degrees maximum at any operating frequency.

11. Shutter current shall be measured at rated voltage and 60 seconds after application of this voltage.

12. Apply 28 Vdc pulses to the shutter circuit. Each time the shutters open, a low-level signal is transmitted through the tube and is registered on a counting circuit. At the end of this test, the number of pulses received should equal the number of pulses applied to the shutter.

13. The shutter shall be held open during vibration by the specified minimum voltage. Vibrate in a plane parallel to the axis of the shutter. The insertion loss test set up shall be used with the unit under test mounted on the vibrator such that a standard waveguide section of similar geometry may be substituted for the shutter tube. The modulation (ΔLi) due to vibration shall not be greater with the shutter tube in the line than with the waveguide section.

14. Shutter phase modulation.

    (a) Introduction. The measurement of rf phase modulation introduced during vibration of the shutter mechanisms in crystal protector tubes can be performed by either of the two indirect techniques to be described. Direct measurement of this parameter is not practical because the phase modulation introduced by the waveguide test equipment will generally be far greater than the permissible shutter modulation.

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NOTES: -Continued

14. Shutter phase modulation. -Continued
(b) Descriptions of shutter mechanism.

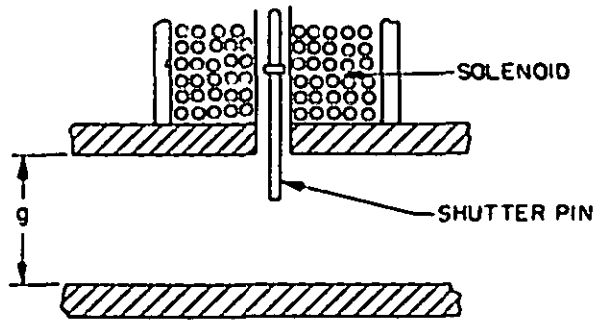


FIGURE 1. Sketch of shutter mechanism.

(1) The shutter pin is attached to a magnetic armature which is located in the magnetic field set up by a solenoid. In the absence of solenoid voltage, the shutter pin is forced against the opposing wall by the force of a shutter closing spring. The rf gap "g" is shorted in this condition and reflects incident signals. With the application of solenoid voltage, the shuter pin is withdrawn from the rf gap against the force of the spring and comes to rest at a height "h" above the opposing wall. Under this condition, the shutter mechanism introduces no appreciable loss to rf signals passing through the crystal protector tube.

(c) Phase calibration.

(1) A shutter mechanism consisting of the shutter pin and the two opposing walls is assembled into a flanged section of waveguide. A calibrated micrometer drive is attached to the shutter pin and permits manual drive of the pin from the shorted condition (i.e., shutter pin resting against the opposing wall) to the "open" condition (i.e., shutter pin at a distance "h" from the opposing wall).

(2) The waveguide section containing the micrometer driven shutter assembly is inserted into an rf phase bridge as shown on figure 2.

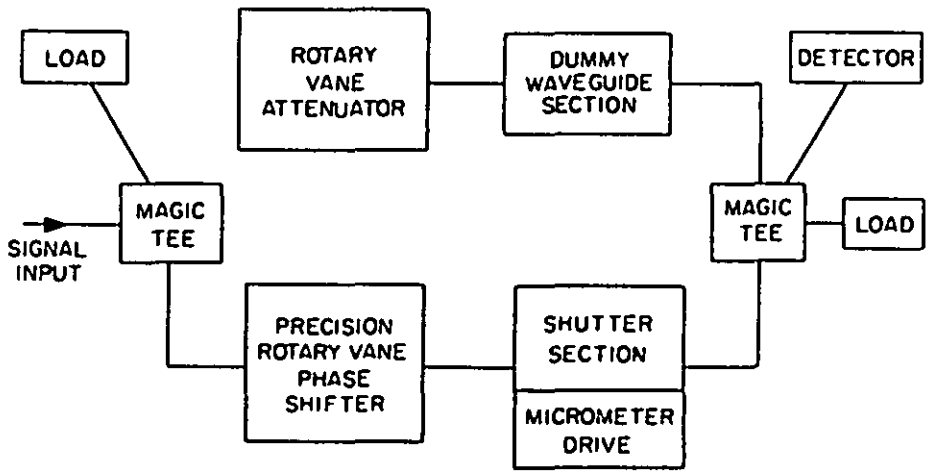


FIGURE 2. Shutter phase calibration.

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NOTES: -Continued

14. Shutter phase modulation. -Continued
    (c) Phase calibration. -Continued
        (3) Using the micrometer drive, the shutter pin is driven against the opposing wall and the phase bridge is balanced with the phase shifter. A balanced condition, as indicated by a null at the detector output, will require insertion of attenuation in the attenuator leg to compensate for the isolation of the shutter. An exact balance is not required at this point. The micrometer reading for this shorted condition of the shutter is recorded.
        (4) The shutter pin is then retracted from the shorted condition in small increments and the phase bridge rebalanced at each micrometer setting. Both the attenuator and the phase shifter shall be reset at each balancing in order to obtain the maximum depth of null at the detector. It will be noted that the amount of phase shifter change required per increment of micrometer movement will diminish as the shutter pin moves out of the gap "g".
        (5) Further retraction of the shutter pin outside the gap region will lead to a critical separation "d", beyond which the motion of the shutter pin no longer has any effect on the phase characteristic of the assembly. This condition will be evidenced by the fact that the bridge will remain in balance (null at the detector) without requiring further adjustment of the phase shifter. The micrometer reading at the critical separation "d" is then subtracted from the initial reading at the shorted condition to establish the value of "d".
    (d) Vibration testing. Vibration testing shall be carried out over the frequency range extending from 10 to 500 to 10 Hz at a 0.06-inch double amplitude (to a maximum of 10 G) per test condition A, method 204 of MIL-STD-202. In order to avoid phase modulation, the shutter pin separation from the opposing wall shall always be greater than the critical spacing "d" for all conditions of vibration. Either optical or electrical means may be used to demonstrate that the shutter design has satisfied this requirement. The voltage applied to the shutter solenoid shall be the minimum allowable under the applicable specification (21 Vdc). The shutter shall be vibration tested at the minimum shutter holding power level determined under quiescent conditions at Es = 21 Vdc and TA = +125° C.
    (e) Optical measurement technique. The wall surrounding the shutter pin is cut back or removed entirely in order to expose the shutter pin for optical observation. The shutter mechanism is then mounted on a suitable vibration table and the shutter pin is observed through a rigidly mounted telescope. The shutter is illuminated by a strobe light which is synchronized to the motion of the vibration table. The telescope is aimed so that a reference marker (e.g., crosspairs) is pointed at the critical distance "d" from the wall opposing the shutter pin. The shutter pin is then withdrawn from the opposing wall by application of the reduced shutter voltage and the assembly is vibrated over the applicable range. Any travel of the shuter pin from its fully withdrawn position (distance "h" from the opposing wall) is observed through the telescope. The extent of this travel shall not reach beyond the critical spacing "d".
    (f) Electrical measurement technique. For this technique, the opposing wall is drilled out similarly to the wall containing the shutter pin. An insulated electrical contact is then passed through this wall and adjusted until its tip is at the distance "d" above the end of the shutter opposing wall. The wall surrounding the shutter pin is cut back or removed as for the optical technique. The electrical connections are then made as shown on figure 3. The shutter pin is maintained at case potential through the shutter closing spring.

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NOTES: -Continued

14. Shutter phase modulation. -Continued.

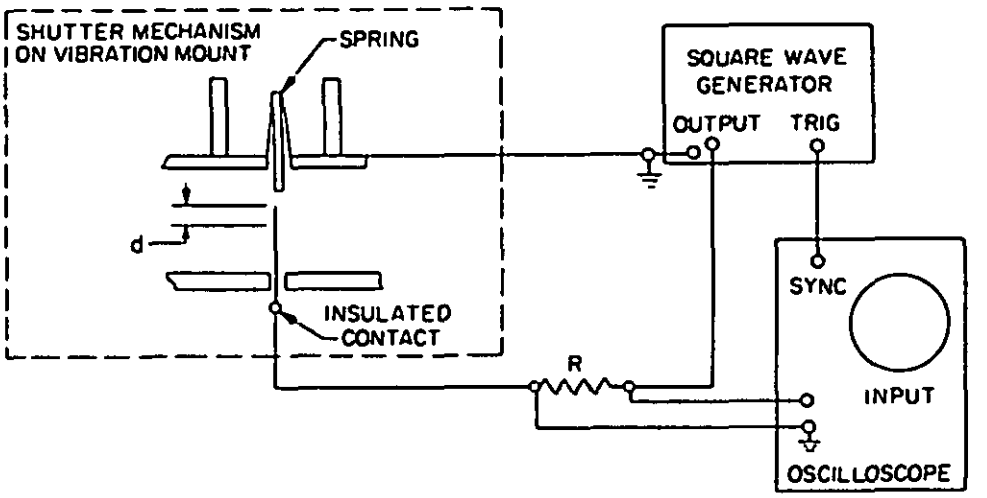


FIGURE 3. Test setup for electrical measurement shutter phase modulation.

(g) The shutter is now vibrated through the specified range. Contact with the insulated electrical contact is made whenever the shutter pin travel comes within the distance "d" of the opposing wall. Current will then flow through the resistor "R" and the monitoring oscilloscope will display the waveform of the square-wave generator under vibration.

15. The current required to hold the shuter open shall be within the specified limit when the tube is subjected to the following shock test. The tubes shall sustain 18-impact shocks (three in each direction of the three mutually perpendicular axes) of 50 G, each shock having a duration of 11 ±1 ms. The shutter shall be capable of meeting this requirement over the temperature range extending from -55° to +125° C.

16. The tube shall be terminated with a type 1N78 crystal mounted in a crystal holder and be provided with a dc ground return of 100 ±10 ohms. At the beginning of the life test, the crystal noise figure (NF) shall meet the minimum requirements for the type 1N78 crystal under the standard specified conditions. The initial crystal noise figure shall be recorded.

17. The type 1N78 crystal noise figure degradation from the initial value shall not exceed the specified limit. One crystal failure will be allowed during the life test. If a crystal failure occurs, a new crystal shall be inserted in the mount, and the life test shall continue for a minimum of 100 hours.

18. During the breakdown test from +25° C breakdown to +125° C breakdown, the change in power required shall not be greater than -30 mw to +60 mw at any selected frequency, and when returned to room temperature reading will return to original reading within ±20 mw.

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NOTES: -Continued

19. The ignitor current noise test is performed in a circuit setup as shown on figure 4. With rated voltages applied to the shutter and to the ignitor, adjust the CW signal from the signal generator to 16.2 GHz. By means of the attenuator, set the crystal current as shown on the meter to 0.5 to 0.6 milliamperes. Set scope internal sweep for 50 μs/cm and vertical (type H plug-in) for 5 mv/cm, or higher. Vary signal frequency slowly from 15.8 to 17.0 GHz. If tube under test is noisy, high amplitude, sharp spikes about 10 μs apart will appear on the scope above the "grass" level. If the spikes can be made to disappear by turning the keep-alive voltage off, the tube is noisy and should be rejected.

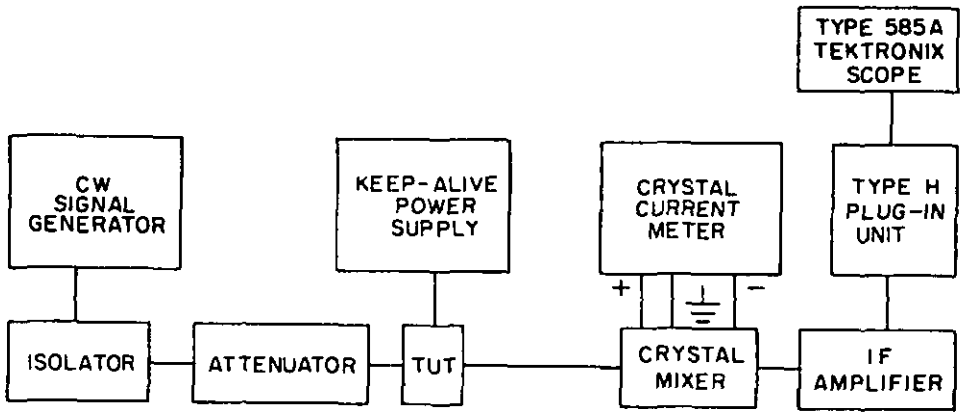


FIGURE 4. Ignitor current noise test setup.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

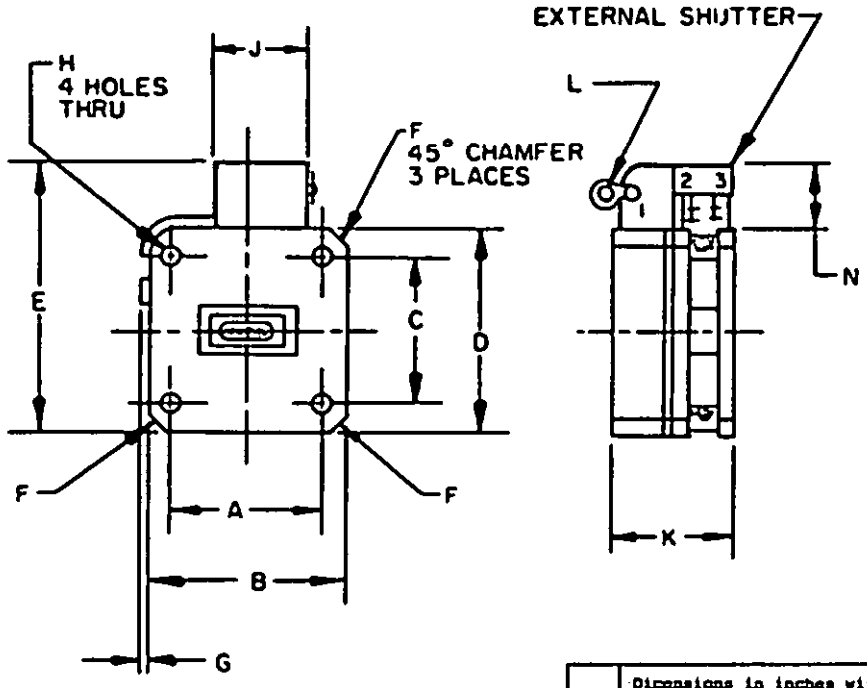
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

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INPUT SIDE

NOTES:
a. Exposed length of shutter post terminals (2) shall be .090(2.29 mm) min and shall be free of potting and paint.
b. Outside surfaces, except mounting surfaces of flanges to have one coat zinc chromate primer per MIL-P-8585 plus two coats of MIL-E-5556 camouflage dull black enamel AMA 604 or FED-STD-595 color 37030. Application of finish shall be in strict accordance with MIL-F-7179 and MIL-F-18264.
c. Marking to be stamped with white ink at locations shown in accordance with the requirements of MIL-STD-130.
d. Exhaust tube, tuning screws, and other elements not to extend beyond flange outlines, except where shown.
e. Surface shall be insulated to withstand 50 Vdc minimum when applied.
f. Shutters shall be sealed.
g. No wires are to be connected to the TR tube terminals. Two orange wires 10-inches long, stripped and tinned shall accompany each TR tube in the shipping container.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note 3)		
G		.125 (3.18)
H	.144dia(3.66)	.147dia(3.73)
K	.803 (20.40)	.805 (20.45)
Quality conformance inspection, part 2		
N		.500 (12.70)
L		.093dia(2.36)
Periodic-check tests		
A	.990 (25.15)	1.000 (25.40)
B	1.297 (32.94)	1.327 (33.71)
C	.951 (24.16)	.961 (24.41)
D	1.297 (32.94)	1.327 (33.71)
E		1.812 (46.02)
F	.035 (.89)	.065 (1.65)
J		.750 (19.05)

FIGURE 5. Outline drawing of electron tube type DOD-005.

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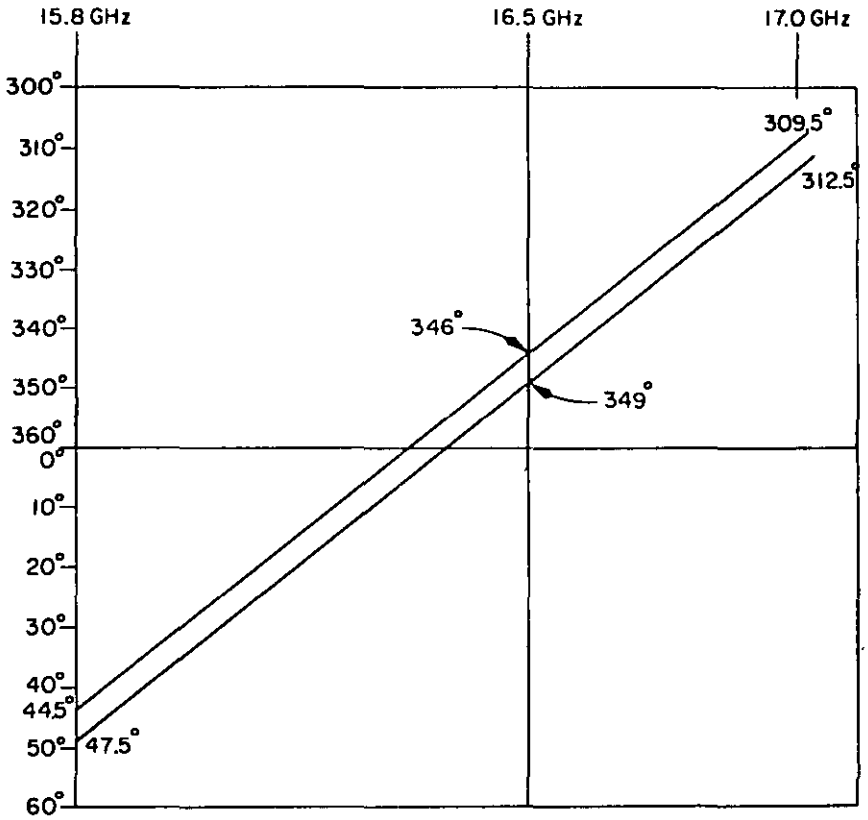


FIGURE 6. Frequency versus degrees phase shift.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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NAVAL ELECTRONIC SYSTEMS COMMAND
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